



PH 422: Statistical Mechanics

Course Code	PH 422
Course Name	Statistical Mechanics
L-T-P-C	4-0-0-4
Prerequisites	Undergraduate Physics Courses and First course Quantum mechanics and faculty consent.
Intended for	I-Ph.D, M.Sc., B.Tech 3rd and 4th Year
Distribution	Elective B.Tech/ Core I-Ph.D
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Contents

- **Module I**
 - Probability concepts and examples - random walk problem in one dimension mean values probability distribution for large N. Probability distribution of many variables. [6 Lectures]
- **Module II**
 - Liouville equation- Boltzmann transport equation, Gibbsian ensemble, BBGKY hierarchy, Boltzmann's H-theorem, Maxwell-Boltzmann distribution, Poincare cycle. Phase space and connection between mechanics and statistical mechanics - Microcanonical ensemble. Classical ideal gas. Gibbs's paradox. [12 Lectures]
- **Module III**
 - Canonical ensemble partition function. Helmholtz free energy, Thermodynamics from the partition function. Classical ideal gas- equipartition and virial theorem. Examples Grand canonical ensemble- density and energy fluctuations- Gibbs free energy. [12 Lectures]
- **Module IV**
 - Formulation of quantum statistical mechanics density matrix- micro-canonical, canonical and grand canonical ensembles- Maxwell-Boltzmann, Fermi-Dirac, and Bose-Einstein statistics - comparison Ideal gas in classical and quantum ensembles Ideal Bose systems Examples of quantum ideal gases, Landau diamagnetism, Pauli paramagnetism, quantum Hall effect, phonons in solids, Bose-Einstein condensation, super fluids. [14 Lectures]
- **Module V**
 - Interacting systems: One dimensional and two dimensional Ising models, lattice gas and binary alloy, Bragg-Williams's approximations. [10 Lectures]

Text Books

1. R K Pathria, **Statistical Mechanics**, 3rd Revised Edition, Academic Press Inc., 2011.
2. K Huang, **Statistical Physics**, 2nd Edition Wiley, 2008.

References

1. F. Reif, **Fundamentals of statistical and thermal physics**, Waveland Press, 2010.
2. L D Landau and E M Lifshitz, **Statistical Physics Part I**, 3rd Edition, Butterworth-Heinemann, 2013.
3. Mehran Kardar, **Statistical physics of particles**, Cambridge University Press, 2007.
4. R. C Tolman, **The principles of Statistical Mechanics**, New Edition, Dover Publications Inc., 1980.